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Convergence Criteria of Iterative Methods Based on Landweber Iteration for Solving Nonlinear Problems*

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Landweber iteration $x_{k+1} = x_k - F'(x_k)^*(F(x_k) - y)$ for the solution of a nonlinear operator equation $F(x_0) = y_0$ can be viewed as a fixed point iteration with fixed point operator $x - F'(x)^*(F(x) - y)$. Especially for nonlinear ill-posed problems, it seems impossible to verify that this fixed point operator is of contractive type, which is a typical assumption for proving (weak) convergence of fixed point iteration schemes. However, for specific examples of nonlinear ill-posed problems it is possible to verify conditions of quasi-contractive type. Weak convergence of Landweber iteration can be proven by application of general results for fixed point iterations, based on quasi-contractive type conditions. In a recent paper by Hanke *et al.* a condition on the operator F has been investigated, which guarantees convergence of the Landweber's method. A geometrical interpretation of this condition is given and is compared with well-known conditions in the theory of fixed point iterations. © 1995 Academic Press, Inc.

1. INTRODUCTION

In [1, 3] the solution of a nonlinear (operator) equation

$$F(x) = y, \tag{1.1}$$

where $F: \mathcal{D}(F) \to Y$ with domain $\mathcal{D}(F) \subseteq X$, by the method of Landweber iteration was considered. This method is defined by the iterative scheme

$$x_{k+1} = x_k - F'(x_k)^*(F(x_k) - y). \tag{1.2}$$

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We restrict our attention to Hilbert spaces X, Y with inner products and norms, denoted by $(\cdot, \cdot)$ and $\|\cdot\|$, respectively; they can always be identified from the context in which they appear. Here, $F'(\cdot)$ and $F'(\cdot)^*$ denote the Fréchet-derivative and its adjoint, respectively.

In the sequel it is assumed that

1. Eq. (1.1) has at least one solution x_*, i.e., $F(x_*) = y$;

2. the Fréchet-derivative $F'(\cdot)$ of F is locally uniformly bounded in a neighborhood of x_*.

There are several ways to prove convergence of (1.2). One could consider (1.2) as a fixed point iteration

$$x_{k+1} = U(x_k), \tag{1.3}$$

with fixed point operator

$$U(x) = x - F'(x)^*(F(x) - y). \tag{1.4}$$

In [2] convergence of fixed point iterations (1.3) has been studied under the assumptions that U is pseudocontractive, strictly pseudocontractive, contractive, or strictly contractive. Especially in the practical important case of nonlinear ill-posed problems (1.1) U (in (1.4)) is not strictly contractive, and we think that it is almost impossible to verify any condition of contractivity type. However, in Section 2 conditions of quasi-contactive type (see, e.g., [4, 6, 7]) are used to apply general results for fixed point iterations to (1.3) with fixed point operator (1.4).

In [3], we required

$$\|F(\tilde{x}) - F(x) - F'(x)(\tilde{x} - x)\| \leq \eta \|F(\tilde{x}) - F(x)\| \tag{1.5}$$

to hold locally for $x, \tilde{x} \in \overline{\mathcal{B}_\rho(x_0)}$. If $\|F'(x)\| < 1$ for $x \in \overline{\mathcal{B}_\rho(x_0)}$ and if (1.5) is satisfied with $\eta \leq \frac{1}{2}$, then these conditions are strong enough to ensure at least local convergence of the iterates to a solution x_* of (1.1) in $\mathcal{B}_{\rho/2}(x_0)$ [3]. Moreover, under the same conditions also for a steepest descent method and a two-level algorithm [5] convergence has been proven.

On the first glimpse (1.5) looks like a standard closeness condition—in [3], a comparison with the usual Fréchet-derivative has been discussed. In Section 3 we give a geometrical interpretation of (1.5) and characterize it by a comparison with conditions of quasi-contractive type. Moreover, we show using two finite dimensional problems the importance of conditions of type (1.5) for convergence of (1.2) in practical situations.

We are mainly interested in the solution of nonlinear, infinite dimen-

sional, ill-posed problems by Landweber iteration. If we have only approximate data y^δ with

$$\|y^\delta - y\| \le \delta \tag{1.6}$$

available, then the iterates x_k^δ from

$$x_{k+1}^\delta = x_k^\delta - F'(x_k^\delta)^*(F(x_k^\delta) - y^\delta) \tag{1.7}$$

can converge only if y^δ belongs to range of F; if y^δ does not belong to the range of F, then a stable approximation of x_* can be provided if the iteration is stopped after $k_* = k_*(\delta)$ steps according to a generalized discrepancy principle, i.e.,

$$\|y^\delta - F(x_{k_*}^\delta)\| \le \tau\delta < \|y^\delta - F(x_k^\delta)\|, \qquad 0 \le k < k_*, \tag{1.8}$$

where τ is a positive number depending on η and ψ, satisfying

$$\|F'(x)\| \le \psi, \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{1.9}$$

such that

$$\frac{2(1+\eta)}{\tau} + 2\eta - 2 + \psi^2 < 0. \tag{1.10}$$

In Section 5 we will discuss convergence rates of the Landweber iterates.

2. Convergence Criteria for Fixed Point Iterations of Nonlinear Mappings

In the theory of nonlinear fixed point iterations it is typically assumed that the fixed point operator is of (quasi-) contractive type (see, e.g., [2, 4, 6, 7]):

Definition 2.1. Let $\emptyset \ne \mathcal{F} \subseteq C$ and U: $C \subset X \to X$.
$P_0^\dagger := \{U$: U is $\mathcal{F}$-quasi-strictly contractive in $C\}$: i.e., there exists a constant k with $k < 1$ such that

$$\|U(x) - U(\tilde{x}_*)\| \le k\,\|x - \tilde{x}_*\| \qquad \text{for all } x \in C,\ \tilde{x}_* \in \mathcal{F};$$

$P_1^\dagger := \{U$: U is $\mathcal{F}$-quasi-contractive (or $\mathcal{F}$-quasi-nonexpansive) in $C\}$: i.e.,

$$\|U(x) - U(\tilde{x}_*)\| \leq \|x - \tilde{x}_*\| \qquad \text{for all } x \in C, \tilde{x}_* \in \mathcal{F};$$

$P_2^{\mathcal{F}} := \{U\colon U$ is $\mathcal{F}$-quasi-strictly pseudocontractive in $C\}$: i.e., there exists a constant $k < 1$ such that

$$\|U(x) - U(\tilde{x}_*)\|^2 \leq \|x - \tilde{x}_*\|^2 + k\,\|(I - U)(x) - (I - U)(\tilde{x}_*)\|^2$$
$$\text{for all } x \in C, \tilde{x}_* \in \mathcal{F};$$

$P_3^{\mathcal{F}} := \{U\colon U$ is $\mathcal{F}$-quasi-pseudocontractive in $C\}$: i.e.,

$$\|U(x) - U(\tilde{x}_*)\|^2 \leq \|x - \tilde{x}_*\|^2 + \|(I - U)(x) - (I - U)(\tilde{x}_*)\|^2$$
$$\text{for all } x \in C, \tilde{x}_* \in \mathcal{F}.$$

Closely related to mappings of quasi-contractive type are mappings of "quasi-monotonicity" type (compare again [2]) $T\colon C \to X$:

DEFINITION 2.2. Let $\emptyset \neq \mathcal{F} \subseteq C$.
$M_0^{\mathcal{F}} := \{T\colon T$ is $\mathcal{F}$-quasi-strongly monotone in $C\}$: i.e., there exists a constant $c > 0$ such that

$$(T(x) - T(\tilde{x}_*), x - \tilde{x}_*) \geq c\,\|x - \tilde{x}_*\|^2 \qquad \text{for all } x \in C, \tilde{x}_* \in \mathcal{F};$$

$M_1^{\mathcal{F}} := \{T\colon T$ is $\mathcal{F}$-quasi-strictly monotone in $C\}$: i.e., there exists a continuous strictly increasing function $c(r)$ on $\{r : r \geq 0\}$ with $c(0) = 0$ such that

$$(T(x) - T(\tilde{x}_*), x - \tilde{x}_*) \geq c\,(\|x - \tilde{x}_*\|) \qquad \text{for all } x \in C, \tilde{x}_* \in \mathcal{F};$$

$M_2^{\mathcal{F}} := \{T\colon T$ is $\mathcal{F}$-quasi-uniformly monotone in $C\}$: i.e., there exists a constant $0 < \alpha < 1$ such that

$$(T(x) - T(\tilde{x}_*), x - \tilde{x}_*) \geq \alpha\,(\|T(x) - T(\tilde{x}_*)\|^2) \qquad \text{for all } x \in C, \tilde{x}_* \in \mathcal{F};$$

$M_3^{\mathcal{F}} := \{T\colon T$ is $\mathcal{F}$-quasi-monotone in $C\}$:

$$(T(x) - T(\tilde{x}_*), x - \tilde{x}_*) \geq 0 \quad \text{for all } x \in C, \tilde{x}_* \in \mathcal{F}.$$

For the sake of completeness we state the definition of Ray (U) given in [2]:

DEFINITION 2.3. A mapping W will be said to lie in the ray from the identity mapping I generated by U (denoted by Ray (U)), if there exists a constant $\lambda > 0$ such that

$$W = I + \lambda(U - I) \qquad \text{or} \qquad W = \lambda U + (1 - \lambda)I. \tag{2.1}$$

If $W \in \mathrm{Ray}(U)$, then U and W have the same fixed points and for $\lambda \leq 1$, W is a convex linear combination of I and U. Furthermore, if $W \in \mathrm{Ray}(U)$, then $U \in \mathrm{Ray}(W)$ and $\mathrm{Ray}(W) = \mathrm{Ray}(U)$.

The basic connections between the $P^{\mathcal{F}}$-classes of operators and the $M^{\mathcal{F}}$-classes are summarized below:

THEOREM 2.4. *Let $P_0^{\mathcal{F}}, \ldots, P_3^{\mathcal{F}}, M_0^{\mathcal{F}}, \ldots, M_3^{\mathcal{F}}$ be as in Definition* 2.1, *Definition* 2.2, *respectively. Then the following connections hold*:

1. *$U \in P_3^{\mathcal{F}}$ if and only if $T = I - U \in M_3^{\mathcal{F}}$;*
2. *$U \in P_2^{\mathcal{F}}$ if and only if $T = I - U \in M_2^{\mathcal{F}}$;*
3. *$U \in P_0^{\mathcal{F}}$ (or $U \in P_1^{\mathcal{F}}$) implies that $T = I - U \in M_2^{\mathcal{F}}$, which is contained in $M_3^{\mathcal{F}}$.*

Proof. Analogous to proof of Theorem 1 in [2]. ∎

Analogously to Theorem 2 in [2], one verifies:

THEOREM 2.5. *$U \in P_2^{\mathcal{F}}$ if and only if there exists an element $W \in \mathrm{Ray}(U)$ such that $W \in P_1^{\mathcal{F}}$.*

Theorem 2.5 states that a mapping U is $\mathcal{F}$-quasi-strictly pseudocontractive in C if and only if there exists an element W in $\mathrm{Ray}(U)$ such that W is $\mathcal{F}$-quasi-contractive in C.

If U is $\mathcal{F}$-quasi-strictly pseudocontractive in C, then analogously to the proof of Theorem 2 in [2] one finds that for any $k \leq \lambda < 1$ the mapping

$$U_\lambda = \lambda I + (1 - \lambda)U$$

has the same fixed points as U in C and that U is $\mathcal{F}$-quasi-contractive in C, and therefore (as we will see below) U_λ admits iterative calculations of its fixed points.

Below we bring three lemmata, which will be used in our further considerations; the proofs of these lemmata can be found in [2] (modulo some minor modifications). The following assumption is required in the sequel:

Assumption 2.6. In the remaining of this section we denote by U: $C \subseteq X \to X$, where C is a bounded, closed, convex subset of X, and by $\mathcal{F}$ the nonempty set of fixed points of U in C.

LEMMA 2.7. *Let U: $C \subseteq X \to X$ be $\mathcal{F}$-quasi-contractive in C. Then, the set of fixed points of U in C is convex and closed.*

For the sake of completeness, we add the definitions of "asymptotically regular" and the "reasonable wanderer" (see [2]):

DEFINITION 2.8. 1. A mapping U of C into C is called *asymptotically regular* at x_0 if and only if

$$\|U^n(x_0) - U^{n+1}(x_0)\| \to 0, \qquad \text{as } n \to \infty.$$

2. A mapping U of C into C is said to be a *reasonable wanderer* at x_0 if and only if its successive steps $x_n = U^n(x_0)$ $(n = 1, 2, 3, \ldots)$ satisfy

$$\sum_{n=0}^{\infty} \|x_{n+1} - x_n\|^2 < \infty. \tag{2.2}$$

LEMMA 2.9. *Let U: $C \subseteq X \to C$ be $\mathfrak{F}$-quasi-contractive in C. For given λ, satisfying $0 < \lambda < 1$, $U_\lambda = \lambda I + (1 - \lambda)U$ is a reasonable wanderer.*

The following theorem shows that the sequence $\{x_n\}$ determined by the process

$$x_n = \lambda x_{n-1} + (1 - \lambda)U(x_{n-1}) = U_\lambda(x_{n-1}), \qquad n = 1, 2, \ldots, \tag{2.3}$$

(where $0 < \lambda < 1$) is at least weakly convergent, if the operator is weakly closed, and $\mathfrak{F}$-quasi-contractive in C.

LEMMA *2.10. Let U: $C \subseteq X \to C$ weakly closed and $\mathfrak{F}$-quasi-contactive in C. For given $0 < \lambda < 1$ and $x_0 \in C$, $x_n := U_\lambda^n(x_0)$ has a weakly convergent subsequence and the limit of every convergent subsequence is a fixed point of U in C.*

Proof. For each $y \in \mathfrak{F}$ and each $n \in \mathbb{N}$,

$$\begin{aligned} \|x_n - y\| &= \|\lambda x_{n-1} + (1 - \lambda)U(x_{n-1}) - y\| \\ &\le \lambda \|x_{n-1} - y\| + (1 - \lambda)\|U(x_{n-1}) - y\| \\ &\le \|x_{n-1} - y\|. \end{aligned} \tag{2.4}$$

By induction, we deduce that $\{x_n\}$ is bounded, and thus it has a weakly convergent subsequence $x_l \rightharpoonup z$. Since by Lemma 2.9, U_λ is a reasonable wanderer,

$$\|x_l - x_{l+1}\| \to 0,$$

and consequently $\|x_l - U_\lambda(x_l)\| \to 0$. Then $U_\lambda(x_l) \rightharpoonup z$. Since U is weakly closed $U_\lambda(z) = z$, and since U and U_λ have the same fixed points, $U(z) = z$. ∎

THEOREM 2.11. *Let U: $C \subseteq X \to C$ be weakly closed and $\mathfrak{F}$-quasi-contractive in C. For given $0 < \lambda < 1$ and $x_0 \in C$,*

$$U_\lambda^n(x_0) \rightharpoonup y \qquad \textit{and} \qquad y \textit{ is a fixed point of } U \textit{ in } C.$$

Proof. Since for any $y \in \mathcal{F}$ by (2.4) $\|x_n - y\| \leq \|x_{n-1} - y\|$, the function

$$g(y) := \lim_{n\to\infty} \|x_n - y\|$$

is well defined on $\mathcal{F}$.

We summarize some basic properties of the function g:

1. the function g is defined on the convex set $\mathcal{F}$ (cf. Lemma 2.7);
2. g is convex;
3. g is continuous.

Since g is both convex and continuous on $\mathcal{F}$, g is lower semi-continuous and quasi-convex (see, e.g., [8]). Since g is lower semi-continuous and quasi-convex, and $\mathcal{F}$ is nonempty (by assumption), convex (by Lemma 2.7), and closed (by Lemma 2.7), we deduce from the extremal principle (see, e.g., [8]) that g attains its minimum d_0 on $\mathcal{F}$.

For each $\varepsilon > 0$, $\mathcal{F}_\varepsilon := \{y: g(y) \leq d_0 + \varepsilon\}$ is closed (by the lower semi-continuity of g), convex (by the quasi-convexity of g), and nonempty (by the assumption that $\mathcal{F} \neq \emptyset$). Hence $\cap_{\varepsilon>0}\mathcal{F}_\varepsilon \neq \emptyset$ (since we already proved that there exists a minimizing element of g in $\mathcal{F}$), and, in fact

$$\bigcap_{\varepsilon>0} \mathcal{F}_\varepsilon = \{y: g(y) = d_0\} =: \mathcal{F}_0.$$

Since $\mathcal{F}_0$ is the intersection of the convex, closed sets $\mathcal{F}_\varepsilon$, it is itself convex and closed. Now one can proceed analogously to the proof of Theorem 8 in [2] to verify that $\mathcal{F}_0$ contains exactly one point, which we will denote by y_0.

The remaining proof will be to verify that $x_n \rightharpoonup y_0$. From Lemma 2.10 it follows that $\{x_n\}$ has a weakly convergent subsequence $x_l \rightharpoonup y$ and y is a fixed point of U. From the definition of g follows

$$\begin{aligned} d_0^2 = g^2(y_0) &= \lim_{l\to\infty} \|x_l - y_0\|^2 \\ &= \lim_{l\to\infty} \|x_l - y + y - y_0\|^2 \\ &= \lim_{l\to\infty} \|x_l - y\|^2 + \|y - y_0\|^2 + 2\lim_{l\to\infty}(x_l - y, y - y_0) \\ &= g^2(y) + \|y - y_0\|^2. \end{aligned}$$

Since $g^2(y) \geq d_0^2$, the last equality implies $\|y - y_0\| \leq 0$. Hence $y = y_0$. ∎

THEOREM 2.12. *Let $U: C \subseteq X \to C$ be weakly closed and $\mathcal{F}$-quasi-strictly pseudocontractive in C. Then, for $k < \gamma < 1$ and $x_0 \in C$,*

$$U_\gamma^n(x_0) \rightharpoonup y \qquad \textit{and } y \textit{ is a fixed point of } U \textit{ in } C.$$

Proof. Due to the remark after Theorem 2.5, for $k \le t < 1$, $U_t = tI + (1 - t)U$ is $\mathcal{F}$-quasi-contractive in C. Hence, by Theorem 2.11, for any given $x_0 \in C$ and any fixed λ with $0 < \lambda < 1$ the sequence

$$\{x_n\} = \{(U_t)_\lambda^n(x_0)\}$$

converges weakly to some fixed point y of U in C, since

$$\begin{aligned}(U_t)_\lambda &= \lambda I + (1 - \lambda)U_t \\ &= (\lambda + (1 - \lambda)t)I + (1 - \lambda)(1 - t)U \\ &= \gamma I + (1 - \gamma)U \\ &= U_\gamma,\end{aligned}$$

where $\gamma = \lambda + (1 - \lambda)t$. Since, for any $k < \gamma < 1$, there exist $\lambda \in (0, 1)$, $t \in (k, 1)$ such that $\gamma = \lambda + (1 - \lambda)t$. The proof is complete. ∎

Theorem 2.12 states that for a $\mathcal{F}$-quasi-strictly pseudocontractive operator the iteration scheme (2.3) is weakly convergent for every fixed λ, satisfying $k < \lambda < 1$.

In Lemma 2.9, Lemma 2.10, and Theorem 2.11 the assumption that U maps C into itself is used to guarantee that the iterates stay in C. In the convergence analysis of iterative methods for the solution of nonlinear fixed point iterations it is typically assumed that the initial guess x_0 is already a "good" approximation for the fixed point y, i.e., $\|x_0 - y\| \le \rho/2$. If U is considered on $C = \overline{\mathcal{B}_\rho(x_0)}$, then it follows by induction from (2.4) that $\|x_n - y\| \le \|x_{n-1} - y\| \le \rho/2$. Consequently by triangle inequality $x_n \in C$. Thus, in this important situation, the assumption that the fixed point operator U maps C into itself can be omitted. In Theorem 2.12 it can be argued analogously that for $\|x_0 - y\| \le \rho/2$ and $C = \overline{\mathcal{B}_\rho(x_0)}$, the assumption U: $C \subseteq X \to C$ can be omitted.

3. Convergence Criteria for Landweber Iteration for Given Exact Data

The convergence analyses for Landweber's method in [1, 3] and for the steepest descent method and the two-level algorithm in [5] are based on condition (1.5). We begin this section by giving a geometrical interpretation of (1.5):

Remark 3.1. Let f: $\mathbb{R} \to \mathbb{R}$ be a continuously differentiable function.

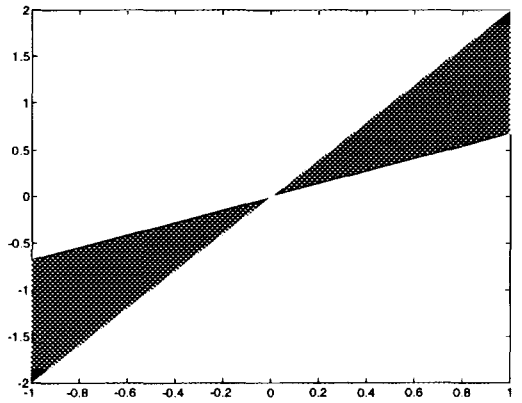


FIG. 3.1. Convex hull of $f(\tilde{x}) + \frac{2}{3}f'(\tilde{x})(x - \tilde{x})$ and $f(\tilde{x}) + 2f'(\tilde{x})(x - \tilde{x})$.

For fixed $\tilde{x} \in \mathbb{R}$ satisfying $f'(\tilde{x}) > 0$ (1.5) with $\eta = 0.5$ means that the graph of f lies entirely between $f(\tilde{x}) + \frac{2}{3}f'(\tilde{x})(x - \tilde{x})$ and $f(\tilde{x}) + 2f'(\tilde{x})(x - \tilde{x})$. For fixed $\tilde{x}$ the convex hull of $f(\tilde{x}) + \frac{2}{3}f'(\tilde{x})(x - \tilde{x})$ and $f(\tilde{x}) + 2f'(\tilde{x})(x - \tilde{x})$ forms a cone with vertex $(\tilde{x}, f(\tilde{x}))$ around the tangent $f(\tilde{x}) + f'(\tilde{x})(x - \tilde{x})$ (see Fig. 3.1). For a continuously differentiable function from $\mathbb{R}$ to $\mathbb{R}$ (1.5) (with $\eta = 0.5$) is satisfied if for every point on the graph of f the graph is contained in the convex hull of $f(\tilde{x}) + \frac{2}{3}f'(\tilde{x})(x - \tilde{x})$ and $f(\tilde{x}) + 2f'(\tilde{x})(x - \tilde{x})$.

For infinite dimensional problems a geometrical interpretation of (1.5) can be given by considering every path from x to $\tilde{x}$ separately.

Motivated by Remark 3.1 we define

DEFINITION 3.2. An operator F satisfies a *uniform tangential cone condition* in $\overline{\mathcal{B}_\rho(x_0)} \subseteq D(F)$ if

$$\begin{aligned} &\|F(\tilde{x}) - F(x) - F'(x)(\tilde{x} - x)\| \le \eta \|F(\tilde{x}) - F(x)\|, \qquad \eta < 1, \\ &\quad x, \tilde{x} \in \overline{\mathcal{B}_\rho(x_0)} \subseteq D(F). \end{aligned} \tag{3.1}$$

As we will see below, (3.1) is strong enough to ensure convergence of (1.2). In order to prove weak convergence of (1.2), we can replace (3.1) by the weaker assumption:

DEFINITION 3.3. Let x_* be a solution of (1.1). An operator F satisfies a *weak tangential cone condition* in $\overline{\mathcal{B}_\rho(x_0)} \subseteq D(F)$ for x_* if

$$\begin{aligned} &(F(x) - F(x_*) - F'(x)(x - x_*), F(x) - F(x_*)) \le \eta \|F(x) - F(x_*)\|^2, \\ &\quad \eta < 1, x \in \overline{\mathcal{B}_\rho(x_0)} \subseteq D(F). \end{aligned} \tag{3.2}$$

On the first glimpse (3.1) and (3.2) look like standard closeness conditions. However, both (3.1) and (3.2) guarantee $\{x_*\}$-quasi-uniform monotonicity of $T(x) = F'(x)^*(F(x) - y)$ in $\overline{\mathcal{B}_\rho(x_0)}$ (and consequently $U(x) = (I - T)(x)$ is $\{x_*\}$-quasi-strictly pseudocontractive in $\overline{\mathcal{B}_\rho(x_0)}$):

THEOREM 3.4. *Let $x_* \in C := \overline{\mathcal{B}_\rho(x_0)}$ be a solution of* (1.1).

1. *If F satisfies a weak tangential cone condition in C for x_* (where η in* (3.2) *can even be replaced by* 1), *then $T(x) = F'(x)^*(F(x) - y)$ is $\{x_*\}$-quasi-monotone for every solution x_* of* (1.1) *in C;*

2. *let F satisfy a weak tangential cone condition in C for x_*. Moreover, let $\|F'(x)\| \le \psi$ on C; then $T(x) = F'(x)^*(F(x) - y)$ is $\{x_*\}$-quasi-uniformly monotone in C.*

Proof. Let x $\in \overline{\mathcal{B}_\rho(x_0)}$; then

$$\begin{aligned}(T(x) - T(x_*), x - x_*) &= (F'(x)^*(F(x) - F(x_*)), x - x_*)\\ &= (F(x) - F(x_*), F'(x)(x - x_*))\\ &= (F(x) - F(x_*), F(x) - F(x_*))\\ &\quad -(F(x) - F(x_*), F(x) - F(x_*) - F'(x)(x - x_*))\\ &\ge (1 - \eta)\|F(x) - F(x_*)\|^2.\end{aligned} \tag{3.3}$$

Thus the $\{x_*\}$-quasi-monotonicity of $T(x)$ is proven. Without loss of generality, we assume $F'(x)^* \neq 0$; then, in the second case

$$\begin{aligned}(1 - \eta)\|F(x) - F(x_*)\|^2 &\ge \frac{1 - \eta}{\|F'(x)^*\|^2}\|F'(x)^*(F(x) - F(x_*))\|^2\\ &\ge \frac{1 - \eta}{\psi^2}\|T(x) - T(x_*)\|^2.\end{aligned} \tag{3.4}$$

Thus, T is $\{x_*\}$-quasi-uniformly monotone in x. ∎

We are especially interested in the solution of nonlinear ill-posed operator equations (1.1): In many nonlinear ill-posed problems the situation occurs that the operator F is compact, weakly closed and $\{F'(x)^*: x \in D(F)\}$ is uniformly bounded: Let $x_n \in \mathcal{D}(F)$, $x_n \rightharpoonup x_*$, and $\|x_n - x_*\| = c \neq 0$. From the compactness, together with the weak closedness of F, it follows that $F(x_n) \rightharpoonup y$, and again from the compactness of F it follows that $F(x_n) \to y$; moreover, due to the uniform boundedness of $\{F'(x)^*: x \in D(F)\}$, $F'(x_n)(x_n - x_*)$ is uniformly bounded in n. Therefore,

$$|(T(x_n) - T(x_*), x_n - x_*)| = |(F(x_n) - y, F'(x_n)(x_n - x_*)| \\ \leq \|F(x_n) - y\| \, \|F'(x_n)(x_n - x_*)\| \to 0.$$

Thus, T is not $\{x_*\}$-quasi-strictly monotone and consequently not $\{x_*\}$-quasi-strongly monotone. This shows that for this important subclass of nonlinear-ill-posed problems the theory of $\{x_*\}$-quasi-strictly monotone (respectively, $\{x_*\}$-quasi-strongly monotone) operators is not applicable.

In the following lemma, we verify that if $T(x) = F'(x)^*(F(x) - y)$ is $\{x_*\}$-quasi-strongly monotone, then F satisfies a weak tangential cone condition for x_* in C:

LEMMA 3.5. *Let $x_* \in C$ be a solution of* (1.1). *Moreover, let $C := \overline{\mathcal{B}_\rho(x_0)} \subseteq D(F)$. If $\|F'(x)^*\| \leq \psi$ for every $x \in C$, then if $T(x) = F'(x)^*(F(x) - y)$ is $\{x_*\}$-quasi-strongly monotone in C, then F satisfies a weak tangential cone condition for x_* in C.*

Proof. From the definition of the $\{x_*\}$-quasi-strong monotonicity of T follows (note that $T(x_*) = 0$)

$$(T(x) - T(x_*), x - x_*) = (F(x) - y, F'(x)(x - x_*)) \geq \alpha \|x - x_*\|^2,$$

and consequently

$$\|F(x) - y\|^2 - (F(x) - y, F(x) - y - F'(x)(x - x_*)) \geq \alpha \|x - x_*\|^2.$$

Since

$$\|F(x) - F(x_*)\| = \left\| \int_0^1 F'(x + t(x_* - x))(x_* - x) \, dt \right\| \leq \psi \|x - x_*\|,$$

where ψ is a bound for $\|F'(x)^*\|$ in C, we get

$$\left(1 - \frac{\alpha}{\psi^2}\right) \|F(x) - y\|^2 \geq (F(x) - y, F(x) - y - F'(x)(x - x_*)).$$

Consequently F satisfies a weak tangential cone condition for x_* in C. ∎

We summarize the previous results in the following remark:

Remark 3.6. Without stating all explicit assumptions on F, we summarize the preceding results: Let x_* be a solution of (1.1) in $C := \overline{\mathcal{B}_\rho(x_0)} \subseteq D(F)$. Then,

T is $\{x_*\}$-quasi-strongly monotone in C
$\Rightarrow F$ satisfies a weak tangential cone condition for x_* in C
$\Rightarrow T$ is $\{x_*\}$-quasi-uniformly monotone in C
$\Rightarrow T$ is $\{x_*\}$-quasi-monotone in C.

Moreover,
T is $\{x_*\}$-quasi-strongly monotone in $C \Rightarrow T$ is $\{x_*\}$-quasi-strictly monotone in C and

F satisfies a uniform tangential cone condition in C
$\Rightarrow F$ satisfies a weak tangential cone condition for $\mathcal{F}$ in C.

We note that we could not have compared $\{x_*\}$-quasi-strict monotonicity of $T(x) = F'(x)^*(F(x) - y)$ with a weak tangential cone condition for x_* of F.

As in the linear case, Landweber iteration can converge only if (1.1) is properly scaled. For our analysis we assume first that (3.2) (with $0 \leq \eta < 1$) and (1.9) hold. Under these assumptions we can show a certain monotonicity of the iteration error:

PROPOSITION 3.7. *Assume that x_* is a solution of* (1.1) *in $\mathcal{B}_{\rho/2}(x_0)$. If* (3.2), (1.9) *hold, and if*

$$2\eta - 2 + \psi^2 < 0, \tag{3.5}$$

then

$$\|x_* - x_{k+1}\| \leq \|x_* - x_k\|, \qquad k = 0, 1, \ldots. \tag{3.6}$$

Moreover,

$$\sum_{k=0}^{\infty} \|y - F(x_k)\|^2 < \infty. \tag{3.7}$$

Proof. From (1.2), (3.2), and (1.9) we obtain

$$\begin{aligned} &\|x_* - x_{k-1}\|^2 - \|x_* - x_k\|^2 \\ &\quad = 2(x_k - x_*, x_{k+1} - x_k) + \|x_{k+1} - x_k\|^2 \\ &\quad = 2(y - F(x_k), F(x_*) - F(x_k) - F'(x_k)(x_* - x_k))) \\ &\quad - (y - F(x_k), (I - F'(x_k)F'(x_k)^*)(y - F(x_k))) - \|y - F(x_k)\|^2 \\ &\quad \leq 2\eta \|y - F(x_k)\|^2 - (2 - \psi^2)\|y - F(x_k)\|^2. \end{aligned} \tag{3.8}$$

Since $2\eta - 2 + \psi^2 < 0$ the right-hand side of (3.8) is negative, and thus

the monotonicity of (3.6) follows. Note that by induction $\|x_0 - x_{k+1}\| \le \|x_0 - x_*\| + \|x_* - x_{k+1}\| < \rho$, and thus in the above inequality (3.2) is applicable. ∎

We note that Proposition 3.7 is applicable for every $\eta < 1$, if (1.1) is properly scaled. For $\eta = 1$ no monotonicity can be proven by the methodology used in the proof of this proposition. If F is weakly closed, then together with Proposition 3.7 weak convergence of (1.2) follows:

THEOREM 3.8. *Assume that the set of solutions of* (1.1) *in* $\mathcal{B}_{\rho/2}(x_0)$ *is nonempty. If* F *is weakly closed, if* (3.2), (1.9), *and* (3.5) *hold, then every subsequence of* $\{x_k\}$ *has a weakly convergent subsequence and the limit of every weakly convergent subsequence is a solution of* (1.1)

Proof. By (3.6), $\{x_k\}$ is bounded. Thus $\{x_k\}$ has a weakly convergent subsequence $x_l \rightharpoonup \bar{x}$. From (3.7) it follows that $F(x_l) \to y$ and by the weak closedness of F it follows that $F(\bar{x}) = y$. ∎

If the set of fixed points of $T(x) = F'(x)^*(F(x) - y)$ is closed and convex, then one can proceed analogously to Theorem 2.11 in proving that $\{x_k\}$ converges weakly to a solution of (1.1) in $\mathcal{B}_{\rho/2}(x_0)$.

If (3.1) is satisfied, then all possible solutions of (1.1) in $\mathcal{B}_\rho(x_0)$ can be characterized.

PROPOSITION 3.9. *If* (3.1) *holds and if* $x_* \in \mathcal{B}_{\rho/2}(x_0)$ *is a solution of* (1.1), *then any other solution* $\tilde{x}_*$ *in* $\mathcal{B}_\rho(x_0)$ *fulfills*

$$x_* - \tilde{x}_* \in \mathcal{N}(F'(x_*)),$$

and vice versa. Here, $\mathcal{N}(\cdot)$ *denotes the null space of an operator.*

Proof. [3]. ∎

If F satisfies a uniform tangential cone condition, then Theorem 2.3 in [3] guarantees that (1.2) is (strongly) convergent. The precise statement is the following [3]:

THEOREM 3.10. *If* (3.1), (1.9), *and* (3.5) *are satisfied and if* (1.1) *is solvable in* $\mathcal{B}_{\rho/2}(x_0)$, *then* $\{x_k\}$ (*the iterates according to* (1.2)) *converge to a solution* $x_* \in \overline{\mathcal{B}_\rho(x_0)}$ *of* (1.1). *If* $x^\dagger$ *denotes the unique solution of minimal distance to* x_0, *and if in addition* $\mathcal{N}(F'(x^\dagger)) \subseteq \mathcal{N}(F'(x))$ *for all* $x \in \overline{\mathcal{B}_\rho(x_0)}$, *then* x_k *converges to* $x^\dagger$.

At the end of this section we summarize (weak) convergence results of Sections 2 and 3 for Landweber's method: With the notation $T(x) = F'(x)^*(F(x) - y)$, $U(x) = I - F(x)$ we proved that if U is weakly closed and U is $\{x_*\}$-quasi-strictly pseudocontractive in the set of fixed points, then $U_\lambda^n(x_0)$ converges weakly to a fixed point of U. A numerical implementation

of the iterative scheme $U_\lambda^n(x_0)$ requires the knowledge of the constant k ($k < \lambda < 1$) in Theorem 2.12, which requires the knowledge of the "degree" of the $\{x_*\}$-quasi-strict pseudocontractivity of the fixed point operator U. From Theorem 3.4 we find that if F satisfies a weak tangential cone condition, then the corresponding fixed point operator is $\{x_*\}$-quasi-strictly pseudocontractive. This assumption is strong enough to guarantee weak convergence of Landweber's method (at least set valued) to a solution of (1.1), and not only to a fixed point of U, i.e., to a critical point. Note that if x_* is a critical point or a solution of (1.1), then it is a fixed point of U. If F satisfies a uniform tangential cone condition, then we get strong convergence of the iteration scheme; moreover, this is again a solution of (1.1) and not only a critical point!

Assumptions on F	Type of convergence	Reference
U is weakly closed, U is $\mathcal{F}$-quasi-strictly pseudocontractive in C	Weak convergence of $U_\lambda^n(x_0)$ to a fixed point of U	Theorem 2.12
F is weakly closed, F satisfies a weak tangential cone condition in $\mathcal{B}_\rho(x_0)$ for x_* in $\mathcal{B}_{\rho/2}(x_0)$	Weak convergence (set valued) of (1.2) to a solution of (1.1)	Theorem 3.8
F is weakly closed, the set of fixed points of U is closed and convex, F satisfies a weak tangential cone condition for x_* in $\mathcal{B}_{\rho/2}(x_0)$	Weak convergence of (1.2) to a solution of (1.1)	Remark following Theorem 3.8
F satisfies a uniform tangential cone condition	Strong convergence of (1.2) to a solution of (1.1)	[3]

4. Modified Landweber's Method and the Case of Inexact Data

In the present section we consider the case that only noisy data y^δ are available, which satisfy (1.6). For our analysis we assume that (3.1) (with $0 \le \eta < 1$), (1.9) hold and that additionally τ, ψ, η satisfy (1.10).

Under these assumptions we can show monotonicity of the itertion error of the iterates according to (1.7):

Proposition 4.1. *Assume that* x_* is a solution of (1.1) *in* $\mathcal{B}_\rho(x_0)$, *and denote by* k_* *the termination index of the iteration according to the stopping rule* (1.8), (1.10) *for the case of perturbed data* y^δ *satisfying* (1.6). *If* (3.1) *and* (1.9) *hold, then*

$$\|x_* - x_{k+1}^\delta\| \le \|x_* - x_k^\delta\|, \qquad 0 \le k < k_*, \tag{4.1}$$

where x_{k+1}^δ is calculated from (1.7).

Proof. Let $0 \le k < k_*$.

$$\begin{aligned}
&\|x_* - x_{k+1}^\delta\|^2 - \|x_* - x_k^\delta\|^2 \\
&\quad = 2(x_k^\delta - x_*, x_{k+1}^\delta - x_k^\delta) + \|x_{k+1}^\delta - x_k^\delta\|^2 \\
&\quad = 2(y^\delta - F(x_k^\delta), F(x_*) - F(x_k^\delta) - F'(x_k^\delta)(x_* - x_k^\delta) + y^\delta - y) \\
&\qquad - (y^\delta - F(x_k^\delta), (I - F'(x_k^\delta)F'(x_k^\delta)^*)(y^\delta - F(x_k^\delta))) \\
&\qquad - \|y^\delta - F(x_k^\delta)\|^2 \\
&\quad \le 2\eta\|y^\delta - F(x_k^\delta)\|\,\|y - F(x_k^\delta)\| + 2\delta\|y^\delta - F(x_k^\delta)\| \\
&\qquad - (2 - \psi^2)\|y^\delta - F(x_k^\delta)\|^2 \\
&\quad \le \left(\frac{2(1+\eta)}{\tau} + 2\eta - 2 + \psi^2\right)\|y^\delta - F(x_k^\delta)\|^2.
\end{aligned} \tag{4.2}$$

Since $(2(1 + \eta)/\tau + 2\eta - 2 + \psi^2) < 0$ the monotonicity of $\|x_k^\delta - x_*\|$ follows. ∎

The proof of Proposition 4.1 fails if we try to use instead of (3.1) the weaker condition (3.2)—to apply the methodology of Proposition 4.1 we have to estimate

$$(y^\delta - y, F(x_*) - F(x_k^\delta) - F'(x_k^\delta)(x_* - x_k^\delta)) \le \delta\eta\|F(x_*) - F(x_k^\delta)\|.$$

The last inequality is valid for all data errors $y^\delta - y$ satisfying (1.6) if and only if

$$\|F(x_*) - F(x_k^\delta) - F'(x_k^\delta)(x_* - x_k^\delta)\| \le \eta\|F(x_*) - F(x_k^\delta)\|.$$

By a careful inspection of the proof of Proposition 4.1 we find that for a modified Landweber iteration.

$$x_{k+1}^\delta = x_k^\delta - G'(x_k^\delta)^*(F(x_k^\delta) - y^\delta), \tag{4.3}$$

where for $x_* \in \mathcal{B}_{\rho/2}(x_0)$, a solution of (1.1), G satisfies

$$\|F(x) - F(x_*) - G'(x)(x - x_*)\| \le \eta\|F(x) - F(x_*)\|, \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{4.4}$$

and

$$\|G'(x)\| \leq \psi, \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{4.5}$$

the errors of the iterates are monotonically decreasing in the sense of Proposition 4.1.

Moreover, if G' satisfies (4.5), and additionally for a solution x_* of (1.1),

$$\|F(x_*) - F(x) - G'(x)(x_* - x)\| \leq \eta \|F(x_*) - F(x)\| + C\delta, \quad x \in \overline{\mathcal{B}_\rho(x_0)}. \tag{4.6}$$

Then, analogously to the proof of Proposition 4.1 one verifies, if

$$2\eta + \psi^2 - 2 + \frac{2(1 + \eta + C)}{\tau} \leq 0,$$

that

$$\|x_{k+1}^\delta - x_*\| \leq \|x_k^\delta - x_*\|,$$

where x_k^δ is calculated from the iterative scheme (4.3).

For $G'(x_k^\delta) = F'(x_k^\delta)$, (4.3) reduces to the original Landweber's method (1.7).

In order to prove strong convergence of the iterative scheme (4.3) by given exact data, we assume that

$$\|F(\tilde{x}) - F(x) - G'(x)(\tilde{x} - x)\| \leq \eta \|F(\tilde{x}) - F(x)\|, \qquad x, \tilde{x} \in \overline{\mathcal{B}_\rho(x_0)}, \tag{4.7}$$

(4.5), and (3.5) hold. In Theorem 2.3 in [3] convergence of iterates of (1.2) was proven for (3.1) with $\eta \leq \frac{1}{2}$ and (1.9) with $\psi = 1$. Analogously to Theorem 2.3 in [3] convergence of (4.3) can be proven:

THEOREM 4.2. *If* (4.7), (4.5), *and* (3.5) *are satisfied and if* (1.1) *is solvable in* $\mathcal{B}_{\rho/2}(x_0)$, *then* $\{x_k\}$ (*the iterates according to* (4.3)) *converges to a solution* $x_* \in \overline{\mathcal{B}_\rho(x_0)}$ *of* (1.1). *If* $x^\dagger$ *denotes the unique solution of minimal distance to* x_0, *and if in addition* $\mathcal{N}(G'(x^\dagger)) \subseteq \mathcal{N}(G'(x))$ *for all* $x \in \overline{\mathcal{B}_\rho(x_0)}$, *then* x_k *converges to* $x^\dagger$.

And in the case of inexact data

THEOREM 4.3. *If* (4.7), (4.5) *are satisfied and if* (1.1) *is solvable in* $\mathcal{B}_{\rho/2}(x_0)$, *then if* y^δ *fulfills* (1.6), *and if the perturbed iteration is stopped with* $k_*(\delta)$ *according to the discrepancy principle* (1.8), (1.10), *then*

$$x_{k_*(\delta)}^{\delta} \to x_*, \qquad \delta \to 0.$$

Proof. Analogous to the proof of Theorem 2.4 in [3]. ∎

Theorem 4.3 states that (4.3) together with the discrepancy principle (1.8), (1.10) is a regularization method, i.e., that (1.7) is stable with respect to data perturbations.

The next example shows that assumptions like (3.1) are necessary for convergence of (1.7), and thus are of practical interest and not useful only in theoretical considerations. In particular, we study the solution of a nonlinear, finite dimensional equation with Landweber iteration.

EXAMPLE 4.4. Let

$$\begin{aligned} f: \mathcal{D}(f) = \left[-\frac{1}{2}, \frac{5}{2}\right] \times \left[-\frac{1}{6}, 2\right] &\to \mathbb{R}, \\ (x, y) &\to c\left[g(x)e^{y/2} + \frac{y^2}{1+y^2}\right], \end{aligned} \tag{4.8}$$

where $g(x) = x^6 - 6.75x^5 + 18.25x^4 - 24.25x^3 - 13.75x^2$, and c is a positive constant such that

$$\left(\frac{\partial f}{\partial x}\right)^2 (x, y) + \left(\frac{\partial f}{\partial y}\right)^2 (x, y) \leq q < 1.$$

Here some properties of f are summarized:

1. $f(0, 0) = 0$, and $f(x, y) > 0$ for $(0, 0) \neq (x, y) \in \mathcal{D}(f)$;
2. for $(x, y) \in \mathcal{D}(f)$, $\nabla f(x, y) = 0$ if and only if $(x, y) = (0, 0)$;

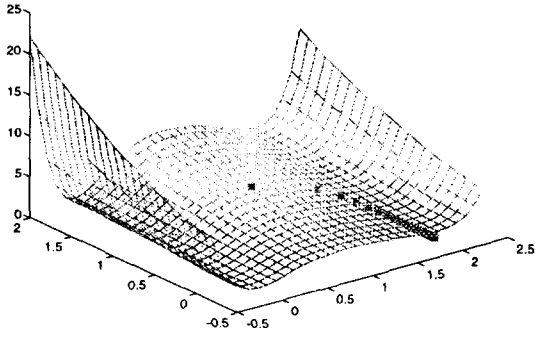


FIG. 4.2. Graph of $f(x, y)$.

3. there exists a point $(x_0, x_0) \in \mathcal{D}(f)$ such that $(\partial f/\partial x)(x_0, x_0) + (\partial f/\partial y)(x_0, x_0) < 0$; for example, the point (1.5, 1.5) satisfies $(\partial f/\partial x)$ (1.5, 1.5) + $(\partial f/\partial y)$ (1.5, 1.5) < 0.

The function f is represented in Fig. 4.2.

We are interested in the solution of

$$f(x) = 0 \tag{4.9}$$

by Landweber's method. Of course the unique solution of (4.9) is given by (0, 0).

Actually we have used in our numerical calculations a projected Landweber's method; i.e., the iterates are projected onto $\mathcal{D}(f)$—as long as the iterates stay in the interior of $\mathcal{D}(f)$, no projection is required.

As an initial guess we used $(x_0, y_0) = (1.5, 1.5)$. At this point $(\partial f/\partial x)$ (1.5, 1.5) + $(\partial f/\partial y)(1.5, 1.5) < 0$; Consequently the next iterate

$$(x_1, y_1) = (x_0, y_0) - \left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0)\right) f(x_0, y_0)$$

is farther away from the unique solution 0 than (x_0, y_0). In Fig. 4.2 we have plotted with stars the Landweber iterates after every 500 iterations. In the plot one sees a hill. The actual solution of our problem (0, 0) is on the left side of the initial guess. But at the initial guess the descending direction is to the right side of the initial guess. Then the iterates are contained in the valley right of the valley where the solution is contained. The iterates move along the valley (right of the actual solution) to the boundary of $\mathcal{D}(f)$, where the projection of the iterates becomes active for the first time. By simple geometrical arguments one finds that no uniform tangential cone condition is satisfied.

In Landweber's method the iterates are in the direction of the gradient of $\frac{1}{2}\|F(x) - y\|^2$, which tends to reduce the square of the residuum. We note that during the iteration process only the gradient of $\frac{1}{2}\|F(x) - y\|^2$ is used, and one can imagine that if the iteration tends (say for x_l) away from the solution, then there is only a little information left in $F'(x)^*(F(x) - y)$ at x_l on the solution. (1.5) guarantees that in the gradient of $\frac{1}{2}\|F(x) - y\|^2$ there is at least as much information on the solution x_* left that the iteration can find its way to the solution; we note that in Proposition 2.1 in [3] it was proven that for any x, $\tilde{x}$ in $\overline{\mathcal{B}_\rho(x_0)}$,

$$\frac{1}{1+\eta}\|F'(x)(x - \tilde{x})\| \leq \|F(x) - F(\tilde{x})\| \leq \frac{1}{1-\eta}\|F'(x)(x - \tilde{x})\|.$$

Thus at every point $x \in \overline{\mathcal{B}_\rho(x_0)}$ in $F'(x)(x - x_*)$ the information on the distance of $F(x)$ to the data y (at least approximate) is available. In particular, as one sees in this example, (1.5) is not applicable to functions with deep valleys. On the other hand, by rescaling this function, one realizes that for Fréchet-differentiable operators there cannot be a convergence result, which guarantees convergence of the Landweber iterates in $\overline{\mathcal{B}_\rho(x_0)}$, under the assumption that ρ and ψ are sufficiently small, where ψ is as in (1.9). In order to guarantee convergence additionally some "monotonicity" is required.

In the next example we indicate how (according to Theorem 4.2) a nonconvergent Landweber's iteration can be modified in order to get a convergent iteration scheme. Essentially the idea is to modify (1.3) by replacing the adjoint of $F'(x_k)$ by an operator that contains more (a priori) information on the solution than $F'(x_k)$.

EXAMPLE 4.5. Let

$$g: [0, \tfrac{5}{2}] \to \mathbb{R},$$
$$x \to c(x^6 - 6.75x^5 + 18.25x^4 - 24.25x^3 + 13.75x^2),$$

where $c = \frac{1}{100}$. We have plotted the graph of g in Fig. 4.3.

We want to apply Landweber's method to solve the nonlinear equation

$$g(x) = g(0.25). \tag{4.10}$$

If exact data ($g(0.25)$) are given, Landweber's method is defined by

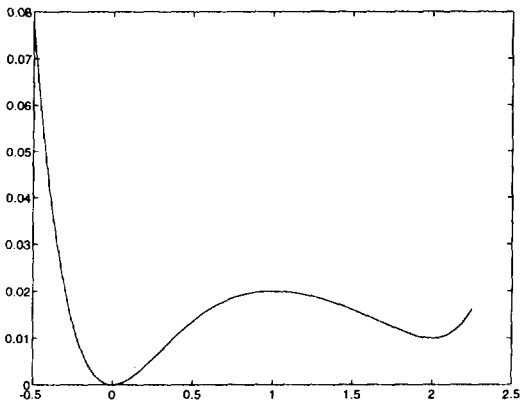


FIG. 4.3. $g(x)$.

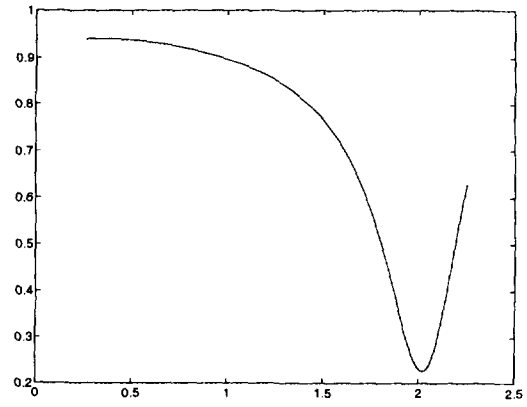


FIG. 4.4. $|g(0.25) - g(x) - 0.2c(0.25 - x)|/|g(0.25) - g(x)|$.

$$x_{k+1} = x_k - g'(x_k)(g(x_k) - g(0.25));$$

as an initial guess we use $x_0 = 2.25$. Since $g'(2) = 0$, (3.1) is not satisfied in any interval containing 0.5 and 2.25. Landweber's method converges to 2.0, and this is certainly no solution of (4.10). In Table I we have summarized the numerical results. Alternatively one could consider the iterative scheme

$$x_{k+1} = x_k - r(g(x_k) - g(0.25)),$$

where r is chosen such that the graph of g between 0.25, 2.25 is contained in a tangential cone with vertex $(0.25, g(0.25))$ (compare Fig. 3.1). In Fig. 4.4 we have plotted $|g(0.25) - g(x) - 0.2c(0.25 - x)|/|g(0.25) - g(x)|$.

TABLE I

Landweber's Method

Iteration number k	x_k
1	2.2493
1,000	2.0787
2,000	2.0391
3,000	2.0210
4,000	2.0115
5,000	2.0064
6,000	2.0036
7,000	2.0020
8,000	2.0012
9,000	2.0007
10,000	2.0004
20,000	2.0000

TABLE II

Modified Method

Iteration number k	x_k
1	2.2496
1,000	2.0387
2,000	1.8804
3,000	1.6595
4,000	1.3017
5,000	0.8297
6,000	0.4684
7,000	0.3208
8,000	0.2728
9,000	0.2574
10,000	0.2524
20,000	0.2500

Consequently, (4.7) is satisfied in this situation for $r = 0.2c$; for this example $|\psi| = 0.2c$, and thus

$$2\eta - 2 + \psi^2 < 0.$$

Thus Theorem 4.2 is applicable and guarantees convergence of the modified Landweber's iterates—note that in this example, x_k is monotonically decreasing to x_*, and thus the technical assumption $\overline{\mathcal{B}_\rho(x_0)} \subseteq \mathcal{D}(F)$ can be weakened to $[x_*, x_0] \subseteq \mathcal{D}(F)$. Of course, for this problem r plays the role of G' in (4.7). In this case the modified Landweber's method converges to 0.25 and thus to a solution of (4.10). In Table II we have summarized the numerical results for the modified Landweber iteration.

5. Convergence Rates Criteria for Ill-Posed Problems

In [3] convergence rates of (1.7) have been considered: In the remainder of this section $x^\dagger$ will denote the unique solution of minimal distance to the initial guess x_0.

Theorem 5.1. *Assume that* (1.1) *has a solution in* $\mathcal{B}_{\rho/2}(x_0)$, *that* y^δ *satisfies* (1.6), *and that* F *fulfills*

$$F'(x) = R_x F'(x^\dagger), \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{5.1}$$

where $\{R_x\colon x \in \overline{\mathcal{B}_\rho(x_0)}\}$ *is a family of bounded linear operators* $R_x\colon Y \to Y$ *with*

$$\|R_x - I\| \le C\|x - x^\dagger\|, \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{5.2}$$

and C *is a positive constant. If*

$$x^\dagger - x_0 = (F'(x^\dagger)^* F'(x^\dagger))^\nu f, \qquad 0 < \nu \le \tfrac{1}{2}, f \in X, \tag{5.3}$$

and $\|f\|$ *sufficiently small, then*

$$\|x^\dagger - x_{k_*}^\delta\| \le c_2 \|f\|^{1/(2\nu+1)} \delta^{2\nu/(2\nu+1)} \tag{5.4}$$

for some constant $c_2 > 0$, *depending on* ν *only.*

Proof. [3]. ∎

It was observed in [3] that for specific examples (4.3) can converge faster than (1.7). These observations are justified by the following theorem:

THEOREM 5.2. *Assume that* (1.1) *has a solution in* $\mathcal{B}_{\rho/2}(x_0)$, *that* y^δ *satisfies* (1.6), *and that* $G'(x)$ *fulfills*

$$G'(x) = R_x G'(x^\dagger), \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{5.5}$$

where $\{R_x\colon x \in \overline{\mathcal{B}_\rho(x_0)}\}$ *is a family of bounded linear operators* $R_x\colon Y \to Y$ *with*

$$\|R_x - I\| \le C\|x - x^\dagger\|, \qquad x \in \overline{\mathcal{B}_\rho(x_0)}, \tag{5.6}$$

and C *is a positive constant. Moreover, we assume that*

$$\|F'(x^\dagger) - G'(x^\dagger)\| = O(\delta).$$

If

$$x^\dagger - x_0 = (F'(x^\dagger)^* F'(x^\dagger))^\nu f, \qquad 0 < \nu \le \tfrac{1}{2},\ f \in X, \tag{5.7}$$

and $\|f\|$ *sufficiently small, then*

$$\|x^\dagger - x_{k_*}^\delta\| \le c_2 \|f\|^{1/(2\nu+1)} \delta^{2\nu/(2\nu+1)} \tag{5.8}$$

for some constant $c_2 > 0$, *depending on* ν *only.*

EXAMPLE 5.3. In [3] we considered the problem of estimating the diffusion coefficient a in

$$\begin{aligned} -(au_s)_s &= f \qquad \text{in } (0,1), \\ u(0) = g_0,\ u(1) &= g_1, \end{aligned} \tag{5.9}$$

where $f \in L^2$ and g_0, g_1 are real numbers. This parameter estimation problem can be formulated as a nonlinear operator equation

$$F(a) = u,$$

where F is defined as the parameter-to-solution mapping

$$\begin{aligned} F\colon \mathcal{D}(F) := \{a \in H^1[0,1]\colon a(s) \ge \underline{a} > 0\} &\to L^2[0,1] \\ a &\to F(a) = u(a); \end{aligned}$$

here $u(a)$ is the solution of (5.9). One can prove that the operator F is Fréchet-differentiable with

$$F'(a)h = A(a)^{-1}[(hu_s(a))_s],$$
$$F'(a)^*w = -B^{-1}[(u_s(a)(A(a)^{-1}w)_s],$$

where $A(a):H^2 \cap H_0^1 \to L^2$ is defined by $A(a)u = -(au_s)_s$, and B denotes the adjoint of the embedding operator from H^2 into L^2. If $u_s(a) \geq \kappa$, $\kappa > 0$, for all $a \in \overline{\mathcal{B}_\rho(a_0)}$, then (3.1) holds locally, which guarantees convergence of the iterated solutions. Unfortunately, F does not fulfill assumptions (5.1), (5.2). To obtain the convergence rate $O(\delta^{1/2})$ under the source condition (5.3) with $\eta = \frac{1}{2}$, we propose the following modified iteration process for this example: instead of using the operator $F'(a_k^\delta)^*$, we suggest using the operator

$$G'(a_k^\delta)^*w := -B^{-1}[u_s^\delta(A(a_k^\delta)^{-1}w)_s].$$

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